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# Tecnologie del Linguaggio Naturale

Parte Prima

## Lezione n. 07

### > Dialogue Systems and ChatBOTs

9 Marzo, 2026

# Outline

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- Introduzione allo studio computazionale del dialogo
- Architetture per ChatBot
  - Eliza Algorithm
- Architetture per Dialog Systems
  - GUS
- Evaluation
  - Trindi Ticklist

# Outline

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- Introduzione allo studio computazionale del dialogo
- Architetture per ChatBot
- Architetture per Dialog Systems
- Evaluation

# What is a dialogue?

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*The rules of conversation are, in general, not to dwell on any one subject, but to pass lightly from one to another without effort and without affectation; to know how to speak about trivial topics as well as serious ones;*

The 18th C. Encyclopedia of Diderot, start of the entry on conversation





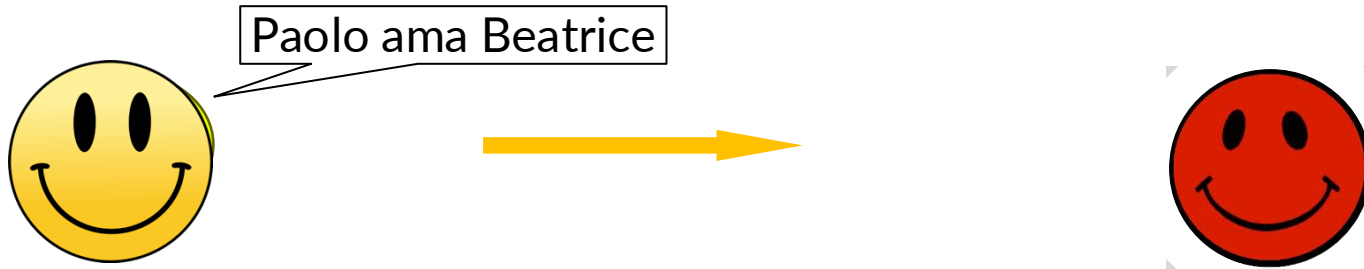
# What is a dialogue?

*A dialogue is a collaborative activity [5]*

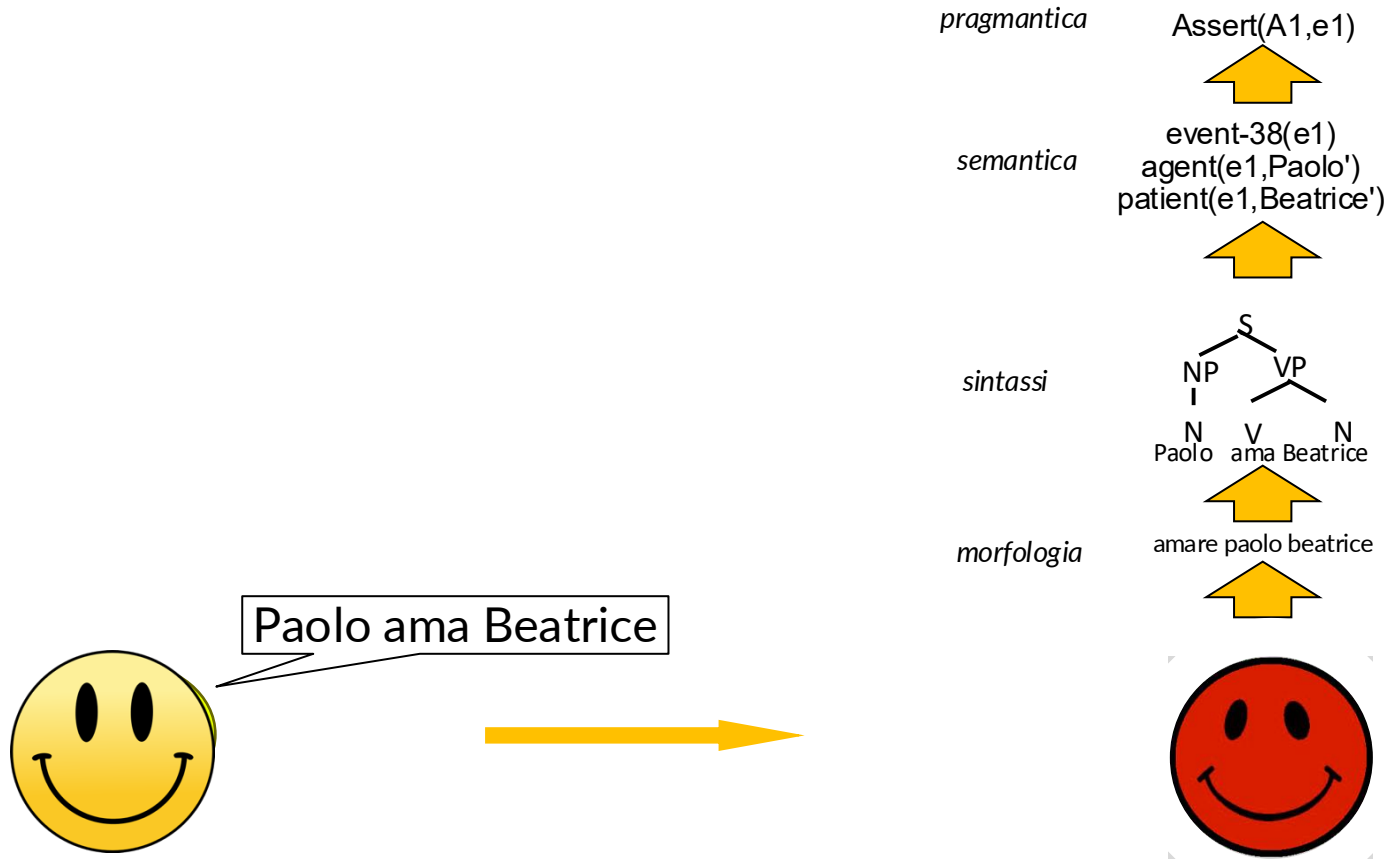
- Activity
  - motivated by the desire to fulfil specific: practical or social goals
  - subject to particular costs to minimise: the communicative effort
  - composed of a temporal sequence of events: the dialogue turns
- Collaborative
  - in order to coordinate their contributions
  - ensure mutual understanding
  - making the interaction more efficient

# Dialogo uomo-uomo (DS)

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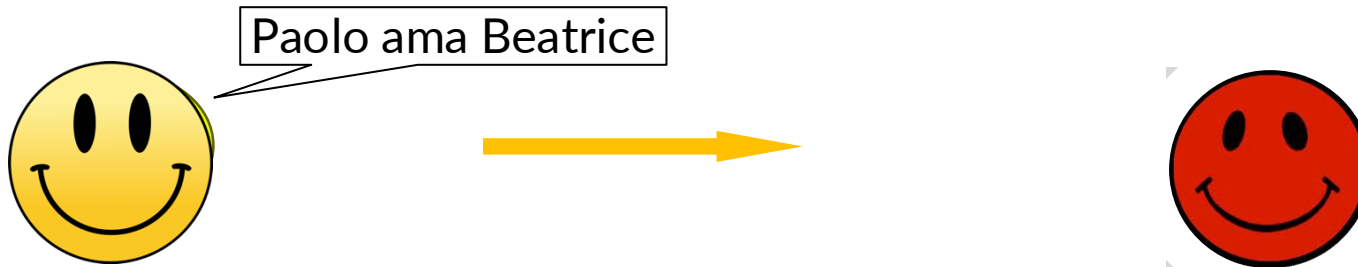
# Dialogo uomo-uomo (DS)



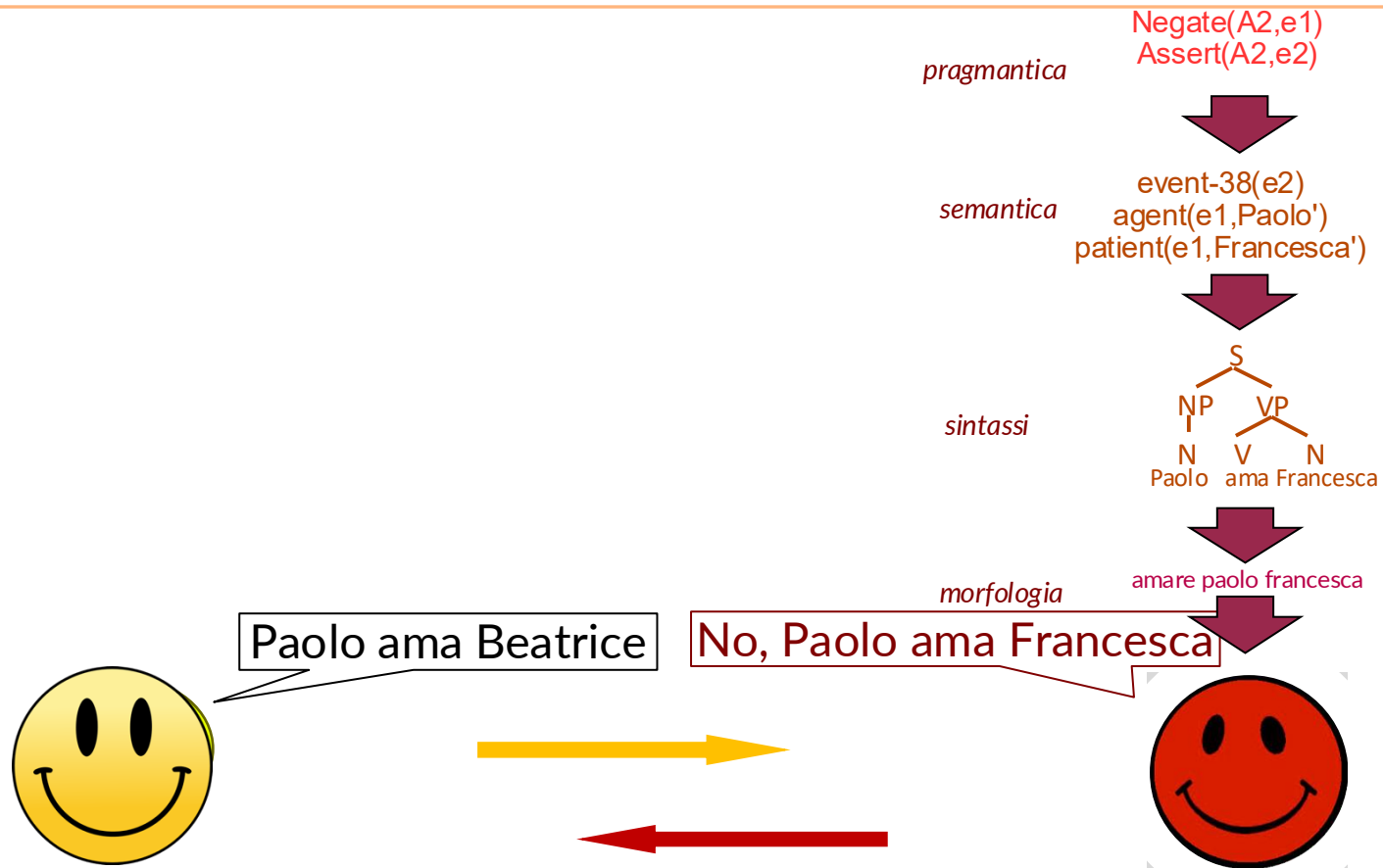
# Dialogo uomo-uomo (DS)

event-38(e1)  
agent(e1,Paolo')  
patient(e1,Beatrice')

???

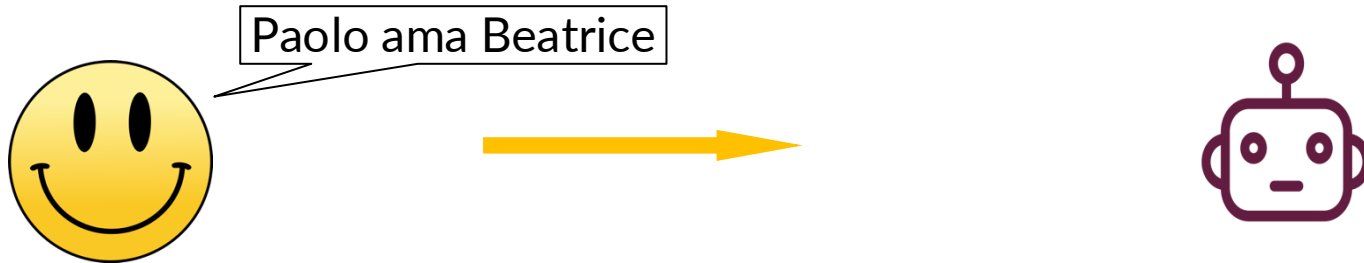


# Dialogo uomo-uomo (DS)



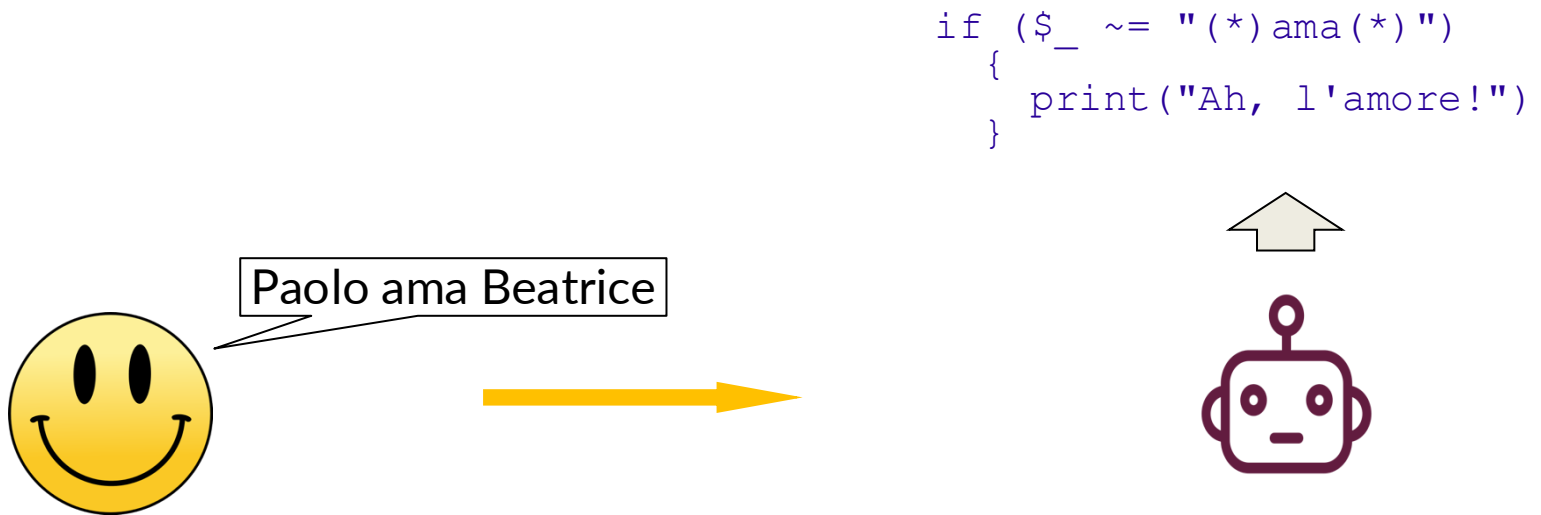
# Dialogo uomo-ChatBot

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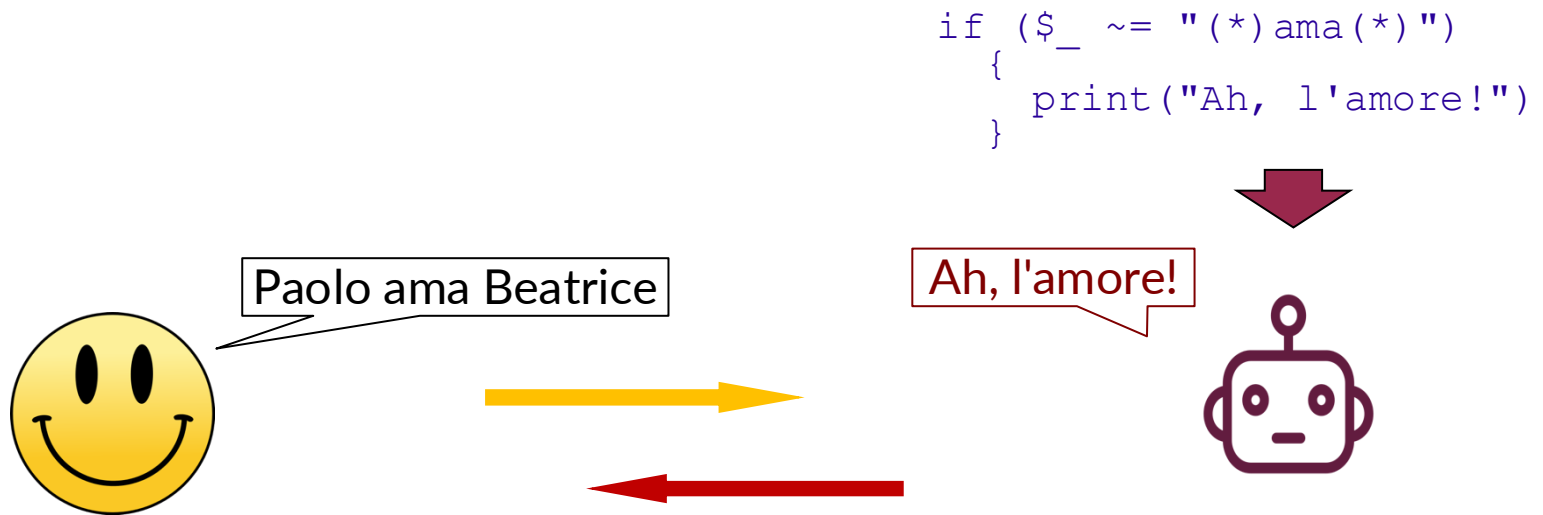


# Dialogo uomo-ChatBot

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# Dialogo uomo-ChatBot





# 6 Key features of human dialogues

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1. Turns
2. Speech acts
3. Grounding
4. Dialogue structure
5. Initiative
6. Implicature

# A telephone conversation between a human travel agent (A) and a human client (C)

C<sub>1</sub>: ... I need to travel in May.  
A<sub>2</sub>: And, what day in May did you want to travel?  
C<sub>3</sub>: OK uh I need to be there for a meeting that's from the 12th to the 15th.  
A<sub>4</sub>: And you're flying into what city?  
C<sub>5</sub>: Seattle.  
A<sub>6</sub>: And what time would you like to leave Pittsburgh?  
C<sub>7</sub>: Uh hmm I don't think there's many options for non-stop.  
A<sub>8</sub>: Right. There's three non-stops today.  
C<sub>9</sub>: What are they?  
A<sub>10</sub>: The first one departs PGH at 10:00am arrives Seattle at 12:05 their time. The second flight departs PGH at 5:55pm, arrives Seattle at 8pm. And the last flight departs PGH at 8:15pm arrives Seattle at 10:28pm.  
C<sub>11</sub>: OK I'll take the 5ish flight on the night before on the 11th.  
A<sub>12</sub>: On the 11th? OK. Departing at 5:55pm arrives Seattle at 8pm, U.S. Air flight 115.  
C<sub>13</sub>: OK.  
A<sub>14</sub>: And you said returning on May 15th?  
C<sub>15</sub>: Uh, yeah, at the end of the day.  
A<sub>16</sub>: OK. There's #two non-stops ... #  
C<sub>17</sub>: #Act... actually #, what day of the week is the 15th?  
A<sub>18</sub>: It's a Friday.  
C<sub>19</sub>: Uh hmm. I would consider staying there an extra day til Sunday.  
A<sub>20</sub>: OK... OK. On Sunday I have ...

# 6 Key features of human dialogues

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1. Turns
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# Turns

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- We call each contribution a "turn"
- As if conversation was the kind of game where everyone takes turns.

# Long and short turns

C<sub>1</sub>: ...I need to travel in May.  
A<sub>2</sub>: And, what day in May did you want to travel?  
C<sub>3</sub>: OK uh I need to be there for a meeting that's from the 12th to the 15th.  
A<sub>4</sub>: And you're flying into what city?  
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A<sub>14</sub>: And you said returning on May 15th?  
C<sub>15</sub>: Uh, yeah, at the end of the day.  
A<sub>16</sub>: OK. There's #two non-stops ...#  
C<sub>17</sub>: #Act... actually #, what day of the week is the 15th?  
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C<sub>19</sub>: Uh hmm. I would consider staying there an extra day til Sunday.  
A<sub>20</sub>: OK... OK. On Sunday I have ...

# Turns

---

- **Turn-taking issues**
  - When to take the floor?
  - When to yield the floor?
- **Interruptions**

# Interruption

C<sub>1</sub>: ...I need to travel in May.  
A<sub>2</sub>: And, what day in May did you want to travel?  
C<sub>3</sub>: OK uh I need to be there for a meeting that's from the 12th to the 15th.  
A<sub>4</sub>: And you're flying into what city?  
C<sub>5</sub>: Seattle.  
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C<sub>19</sub>: Uh hmm. I would consider staying there an extra day til Sunday.  
A<sub>20</sub>: OK...OK. On Sunday I have ...

# Turns

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- **Barge-in**

- Allowing the user to interrupt

- **End-pointing**

- The task for a speech system of deciding whether the user has stopped talking.
  - Very hard, since people often pause in the middle of turns



# 6 Key features of human dialogues

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1. Turns
2. **Speech acts**
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# Speech Acts (aka Dialogue Acts)

Bach and Harnish (1979)

**Constatives:** committing the speaker to something's being the case  
(*answering, claiming, confirming, denying, disagreeing, stating*)

**Directives:** attempts by the speaker to get the addressee to do something  
(*advising, asking, forbidding, inviting, ordering, requesting*)

**Commissives:** committing the speaker to some future course of action  
(*promising, planning, vowing, betting, opposing*)

**Acknowledgments:** express the speaker's attitude regarding the hearer with respect to some social action (*apologizing, greeting, thanking, accepting an acknowledgment*)

# Speech acts

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- "Turn up the music!"

Directive

- "What day in May do you want to travel?"

Directive

- "I need to travel in May"

Constative

- Thanks

Acknowledgement

# 6 Key features of human dialogues

---

1. Turns
2. Speech acts
3. **Grounding**
4. Dialogue structure
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# Grounding

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- Participants in conversation or any joint activity need to establish **common ground**.
- Principle of closure. Agents performing an action require evidence, sufficient for current purposes, that they have succeeded in performing it (Clark 1996, after Norman 1988)
- Speech is an action too! So speakers need to ground each other's utterances.
- **Grounding**: acknowledging that the hearer has understood

# Grounding

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- Grounding is relevant for human-machine interaction
- Why do elevator buttons light up?



# Grounding: Establishing Common Ground

---

A: And you said returning on May 15th?

C: Uh, yeah, at the end of the day.

A: **OK**

C: OK I'll take the 5ish flight on the night before on the 11th.

A: **On the 11th? OK.**

C: ...I need to travel in May.

A: **And**, what day **in May** did you want to travel?

# 6 Key features of human dialogues

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# Conversations have structure

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Local structure between adjacent speech acts, from the field of conversational analysis (Sacks et al. 1974). Called **adjacency pairs**:

- Question ... Answer
- Proposal... Acceptance/Rejection
- Compliments ("Nice jacket!") ... Downplayer ("Oh, this old thing?")

# Conversations have structure

---

## Subdialog for Correction:

- Agent:** OK. There's #two non-stops#

**Client:** #Act- actually#, what day of the week is the 15th?

**Agent:** It's a Friday.

**Client:** Uh hmm. I would consider staying there an extra day til Sunday.

- Agent:** OK...OK. On Sunday I have ...

# Conversations have structure

---

## Subdialogue for clarification:

- User:** What do you have going to UNKNOWN WORD on the 5th?

**System:** Let's see, going where on the 5th?

**User:** Going to Hong Kong.

- System:** OK, here are some flights...

# Conversations have structure

---

## Presequences:

- **User:** Can you make train reservations?
- **System:** Yes, I can.
- **User:** Great, I'd like to reserve a seat on the 4pm train to New York.

# 6 Key features of human dialogues

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# Conversational Initiative

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- Some conversations are controlled by one person
- A reporter interviewing a chef asks questions, and the chef responds. This reporter has the **conversational initiative** (Walker and Whittaker 1990)
- Most human conversations have **mixed initiative**:
  - I lead, then you lead, then I lead.
- **Mixed initiative is very hard** for NLP systems, which often default to simpler styles that can be frustrating for humans:
  - **User initiative** (user asks or commands, system responds)
  - **System initiative** (system asks user questions to fill out a form, user can't change the direction)

# 6 Key features of human dialogues

---

1. Turns
2. Speech acts
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5. Initiative
- 6. Implicature**

# Even harder problems: Inference

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**A:** And, what day in May did you want to travel?

**C:** OK, uh, I need to be there for a meeting that's from the 12th to the 15th.

- Conversational implicature (Grice):
  - maxim of relevance
  - ...



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- Introduzione allo studio computazionale del dialogo
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  - Dialog State and Dialogue Manager
- Evaluation

# Eliza: Weizenbaum (1966)

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Men are all alike.

IN WHAT WAY

They're always bugging us about something or other.

CAN YOU THINK OF A SPECIFIC EXAMPLE

Well, my boyfriend made me come here.

YOUR BOYFRIEND MADE YOU COME HERE

He says I'm depressed much of the time.

I AM SORRY TO HEAR YOU ARE DEPRESSED

...

WHO ELSE IN YOUR FAMILY TAKES CARE OF YOU?

My father

YOUR FATHER

# Eliza: Weizenbaum (1966)

---

You are like my father in some ways.

WHAT RESEMBLANCE DO YOU SEE

You are not very aggressive but I think you don't want me to notice that.

WHAT MAKES YOU THINK I AM NOT AGGRESSIVE

You don't argue with me.

WHY DO YOU THINK I DON'T ARGUE WITH YOU

You are afraid of me.

DOES IT PLEASE YOU TO BELIEVE I'M AFRAID OF YOU

My father is afraid of everybody.

WHAT ELSE COMES TO MIND WHEN YOU THINK OF YOUR FATHER

Bullies.

DOES THAT HAVE ANYTHING TO DO WITH THE FACT THAT YOUR BOYFRIEND MADE YOU COME HERE

# Eliza: Weizenbaum (1966)

---

**Rogerian psychologist:** draw the patient out by reflecting patient's statements back at them. Rare type of conversation in which one can “assume the pose of knowing almost nothing of the real world”

**Patient:** "I went for a long boat ride"

**Psychiatrist:** "Tell me about boats"

# Eliza pattern/transform rules

---

(0 YOU 0 ME)      [*pattern*]

→

(WHAT MAKES YOU THINK I 3 YOU) [*transform*]

0 means Kleene \*

The 3 is the constituent \$i in pattern

You hate me

WHAT MAKES YOU THINK I HATE YOU

# Eliza algorithm

**function** ELIZA GENERATOR(*user sentence*) **returns** *response*

Find the word  $w$  in *sentence* that has the highest keyword rank

**if**  $w$  exists

Choose the highest ranked rule  $r$  for  $w$  that matches *sentence*

$response \leftarrow$  Apply the transform in  $r$  to *sentence*

**if**  $w = \text{'my'}$

$future \leftarrow$  Apply a transformation from the 'memory' rule list to *sentence*

Push *future* onto memory stack

**else** (no keyword applies)

**either**

$response \leftarrow$  Apply the transform for the NONE keyword to *sentence*

**or**

$response \leftarrow$  Pop the top response from the memory stack

**return**(*response*)

# Keywords are ranked from specific to general

I know everybody laughed at me

- “I” is a very general keyword:

I: (I \*) -> (You say you 2)

YOU SAY YOU KNOW EVERYBODY LAUGHED AT YOU

- “Everybody” is much more interesting (someone using universals like everybody/always is probably “referring to some quite specific event or person”)

WHO IN PARTICULAR ARE YOU THINKING OF?

- Implementation: keywords stored with their rank

**Everybody** 5 (*transformation rules*)

**I** 0 (*transformation rules*)

# None patterns (back-up strategy)

---

PLEASE GO ON

THAT'S VERY INTERESTING

I SEE



# Memory

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(MEMORY MY

(0 YOUR 0 = LETS DISCUSS FURTHER WHY YOUR 3)

(0 YOUR 0 = EARLIER YOU SAID YOUR 3)

- Whenever “MY” is highest keyword
  - Randomly select a transform on the MEMORY list
  - Apply to sentence
  - Store on a stack
- Later, if no keyword matches a sentence
  - Return the top of the MEMORY queue instead
- **A hierarchical model of discourse**

# Other Eliza's tricks

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- Rules can refer to classes of words

Family = mother, father, brother, sister  
NOUN = ...

- Don't reuse transforms in the same conversation
  - Whenever we use a transform associated with a pattern, we increment a counter for that rule. So the next time we use the next ranked transform
- Some basic transforms happen during input processing
  - I -> YOU
  - YOU -> I

# Eliza effect

- People became deeply emotionally involved with the program
- Weizenbaum tells the story of his secretary who would ask Weizenbaum to leave the room when she talked with ELIZA
- When he suggested that he might want to store all the ELIZA conversations for later analysis, people immediately pointed out the privacy implications. Suggesting that they were having quite private conversations with ELIZA
- Pareidolia



“Il mio chatbot non mi capisce”.

# Parry

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- Colby 1971 at Stanford
- Same pattern-response structure as Eliza
- But a much richer:
  - control structure
  - language understanding capabilities
  - mental model: Parry has affective variables
    - Anger, Fear, Mistrust
    - “If Anger level is high, respond with hostility”
- The first system to pass the Turing test (in 1971)
  - Psychiatrists couldn’t distinguish interviews with PARRY from (text transcripts of) interviews with real paranoids

# A.L.I.C.E.

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- Artificial Intelligence Markup Language
- Struttura simile a HTML
- Permette di definire delle regole per rispondere all'utente mediante **coppie di adiacenza**

# A.L.I.C.E.: TAG

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- La struttura delle regole si compone di <tag>
- I tag racchiudono altri tag o del testo, secondo un ordine prestabilito

<category>

    <pattern>HI</pattern>

    <template>Hello world!</template>

</category>

# A.L.I.C.E.: Pandorabots

The screenshot displays the Pandorabots web interface. On the left is a dark sidebar with navigation options: MY BOTS (with a plus icon), ci2019 (selected, with a minus icon), Edit, Intents, Deploy, Logs, Delete, DIRECTORY (with a right arrow), and HELP (with a right arrow). The main area features a file explorer on the left with a tree structure: AIML (expanded), udc, test, Maps, Sets, Substitutions, person2, person, normal, gender, denormal, System, and properties. The central editor shows an XML file named 'test' with the following content:

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <aiml version="2.0">
3   <!-- insert your AIML categories here -->
4
5   <category>
6     <pattern>Tell me about *</pattern>
7     <template>https://en.wikipedia.org/wiki/*$1
8   </category>
9
10
11 </aiml>
```

At the top right of the editor is a green 'Publish Changes' button. On the right side, a chat window for 'ci2019' is open, showing a conversation with a robot icon. The chat history includes:

- Robot: I have no answer for that.
- User: ciao
- User: tell me about zebra
- Robot: https://en.wikipedia.org/wiki/zebra
- User: ciao
- Robot: I have no answer for that.

The chat window has a 'Say something ...' input field at the bottom.

# Corpus based chatbots: 2 ways

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## 1. Response by retrieval

- Use information retrieval to grab a response (that is appropriate to the context) from some corpus

## 2. Response by generation

- Use a language model or encoder-decoder to generate the response given the dialogue context



# Corpus-based chatbots require corpora

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- Modern corpus-based chatbots are very data-intensive
- They commonly require hundreds of millions or billions of words

# What conversations to draw on?

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- Transcripts of telephone conversations between volunteers
  - Switchboard corpus of American English telephone conversations
- Movie dialogue
- Hire human crowdworkers to have conversations
  - Topical-Chat 11K crowdsourced conversations on 8 topics
  - EMPATHETICDIALOGUES 25K crowdsourced conversations grounded in a situation where a speaker was feeling a specific emotion
- Pseudo-conversations from public posts on social media
  - Drawn from Twitter, Reddit, Weibo (微博), etc.
  - Tend to be noisy; often used just as pre-training.

# 1. Response by retrieval: classic IR method

---

1. Given a user turn  $q$ , and a training corpus  $C$  of conversation
2. Find in  $C$  the turn  $r$  that is most similar (tf-idf cosine) to  $q$
3. Say  $r$

$$\text{response}(q, C) = \operatorname{argmax}_{r \in C} \frac{q \cdot r}{|q||r|}$$

# 1. Response by retrieval: neural IR method

---

1. Given a user turn  $q$ , and a training corpus  $C$  of conversation
2. Find in  $C$  the turn  $r$  that is most similar (BERT dot product) to  $q$
3. Say  $r$

$$h_q = \text{BERT}_Q(q) [\text{CLS}]$$

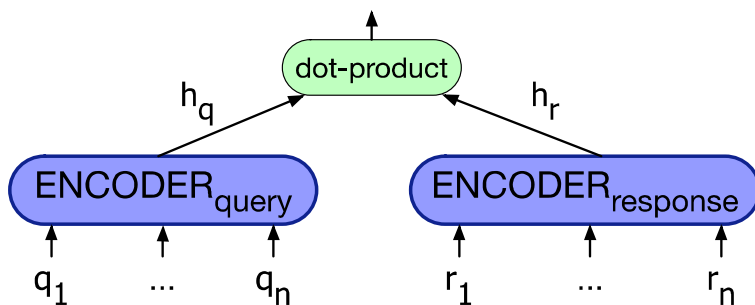
$$h_r = \text{BERT}_R(r) [\text{CLS}]$$

$$\text{response}(q, C) = \underset{r \in C}{\operatorname{argmax}} h_q \cdot h_r$$

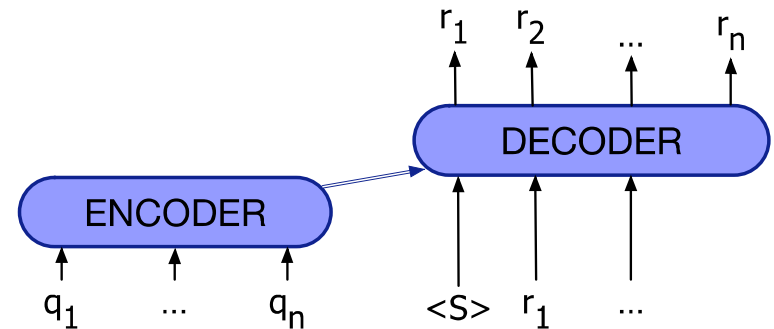
## 2. Response by generation

- Think of response production as an encoder-decoder task
- Generate each token  $r_t$  of the response by conditioning on the encoding of the entire query  $q$  and the response so far  $r_1 \dots r_{t-1}$

$$\hat{r}_t = \operatorname{argmax}_{w \in V} P(w|q, r_1 \dots r_{t-1})$$

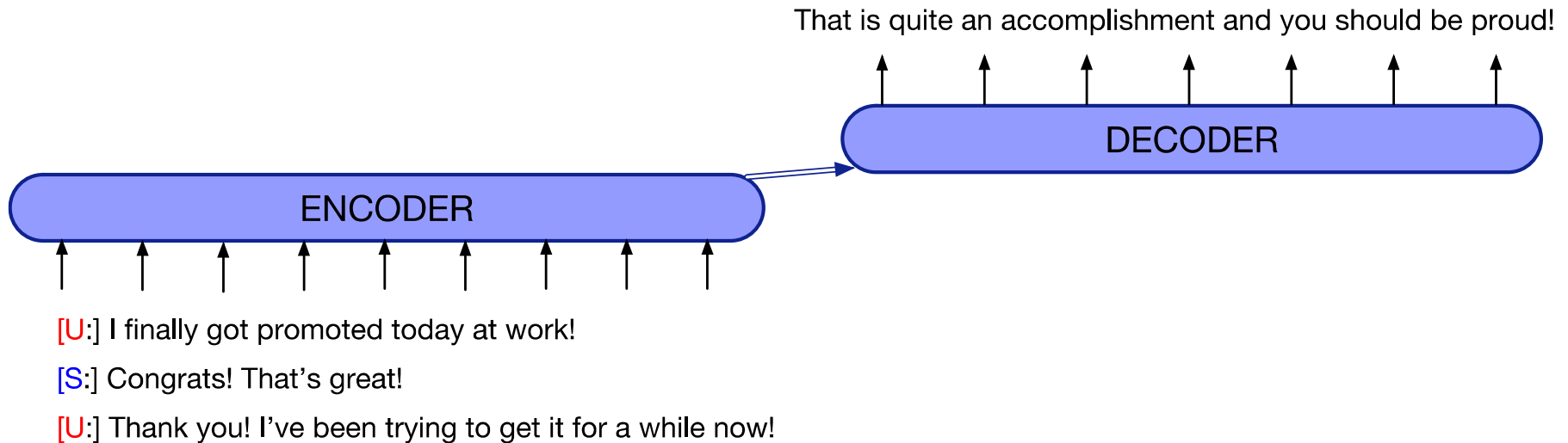


(a) Response by Retrieval



(b) Response by Generation

# Response by generation



# Ongoing research problem: Neural chatbots can get repetitive and boring

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A: Where are you going?

A: How old are you?

B: I'm going to the restroom.

B: I'm 16.

A: See you later.

A: 16?

B: See you later.

B: I don't know what you are talking about.

A: See you later.

A: You don't know what you are saying.

B: See you later.

B: I don't know what you are talking about.

A: You don't know what you are saying.

# Response by retrieving and refining knowledge

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- Can generate responses **from informative text** rather than dialogue
- To respond to turns like “Tell me something about Beijing”
  - Xiaolce collects sentences **from public lectures and news articles**.
  - And searches them using IR based on query expansion from user’s turn
- Can augment encoder-decoder model
  - use **IR to retrieve passages from Wikipedia**
  - concatenate each Wikipedia sentence to the dialogue context with a separator token.
  - Give as encoder context to the encoder-decoder model, which learns to incorporate text into its response



# Hybrid Architectures

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- Chirpy Cardinal (Paranjape et al., 2020) response generation from a series of different generators:
  - GPT-2 finetuned on EmpatheticDialogues
  - GPT-2 finetuned to paraphrase content from Wikipedia
  - Rule-based movie or music generators that produce scripted conversation about a movie or a musician
    - asking the user's opinion about a movie,
    - giving a fun fact,
    - asking the user their opinion on an actor in the movie.

| # | Utterance                                                                                                                                                                                                                      | Entity                     | Commentary                                                                                                                                                                          |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>User:</b> let's chat<br><b>Bot:</b> Hi, this is an Alexa Prize Socialbot. I'd love to get to know you a bit better before we chat! Is it all right if I ask for your name? [ Launch ]                                       | None<br>None               | <b>Launch</b> gives <i>FORCE_START</i> response to begin the conversation.                                                                                                          |
| 2 | <b>User:</b> my name is chris<br><b>Bot:</b> Well it's nice to meet you, Chris! I'm excited to chat with you today. [ Launch ] I hope your afternoon is going well. What are your plans for the rest of today? [ Neural Chat ] | None<br>None               | <b>Launch</b> gives a <i>STRONG_CONTINUE</i> response that needs a prompt, which is supplied by <b>Neural Chat</b> with <i>FORCE_START</i> priority (scripted transition).          |
| 3 | <b>User:</b> hang out with my friends<br><b>Bot:</b> That sounds great, I love spending time with friends. What will you do together? [ Neural Chat ]                                                                          | None<br>None               | <b>Neural Chat</b> generates a GPT-2 response with <i>STRONG_CONTINUE</i> priority.                                                                                                 |
| 4 | <b>User:</b> maybe watch a movie<br><b>Bot:</b> Me too! I love watching movies; I get to learn so much about what the world is like outside of the cloud! Have you seen any movies recently? [ Movies ]                        | None<br>Film               | <b>Movies</b> detects the 'movie' keyword in the user's utterance and gives a <i>FORCE_START</i> response (overriding <b>Neural Chat</b> ), setting current entity to <i>Film</i> . |
| 5 | <b>User:</b> i saw the matrix<br><b>Bot:</b> Nice! Did you like The Matrix? [ Movies ]                                                                                                                                         | The Matrix<br>The Matrix   | The <b>Entity Linker</b> and <b>Entity Tracker</b> identify <i>The Matrix</i> as the current entity.                                                                                |
| 6 | <b>User:</b> i loved it neo is amazing<br><b>Bot:</b> Oooh, yeah, I agree. Hey, isn't Keanu Reeves in that movie? What do you think about Keanu Reeves? [ Movies ]                                                             | The Matrix<br>Keanu Reeves | <b>Movies</b> sets current entity to <i>Keanu Reeves</i> .                                                                                                                          |
| 7 | <b>User:</b> i want to talk about something else                                                                                                                                                                               | None                       | <b>Navigational Intent</b> detects <i>negative intent</i> , so <b>Entity Tracker</b> rejects the current entity.                                                                    |

# Chatbots: pro and con

---

- Pro:
  - Fun
  - Good for narrow, scriptable applications
- Cons:
  - They don't really understand
  - Giving the appearance of understanding may be problematic
  - Rule-based chatbots are **expensive** and **brittle**
  - IR-based chatbots can only mirror training data
- Next steps: integrating chatbot ability into Dialogue Systems

# Outline

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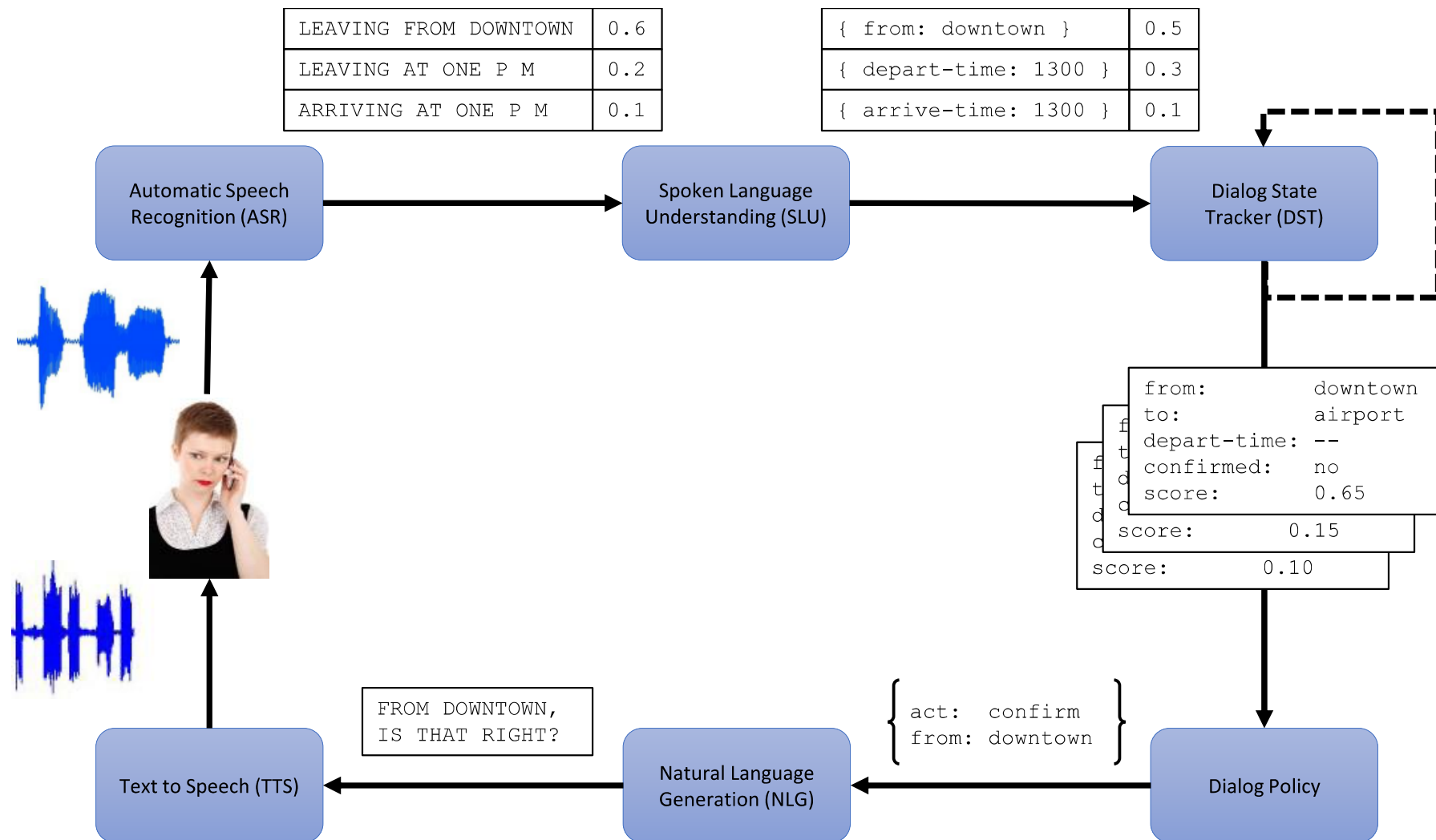
- Introduzione allo studio computazionale del dialogo
- Architetture per ChatBot
- Architetture per Dialog Systems
- Evaluation

# Dialogue Systems applications

<https://www.youtube.com/watch?v=kTcRRaXV-fg>

- Call routing
- Troubleshooting
- Directory assistance
- Telephone-based enquiries, e.g., travel/hotel/car rental enquiries, bank balance
- Desktop information assistance
- Training applications
- Educational and tutoring applications
- Robot control
- Planning assistance
- Information acquisition
- Healthcare systems, e.g., patient monitoring
- Car systems
- Smart home applications
- Ubiquitous applications

# DS Architecture



# Components in a dialogue-state architecture

---

- **NLU:** extracts slot fillers from the user's utterance using machine learning
- **NLG:** produce more natural, less templated utterances
- **Dialogue state tracker:** maintains the current **state of the dialogue** (user's most recent dialogue act, set of slot-filler constraints from user, ...)
- **Dialogue policy:** decides what the system should do or say next
  - **GUS policy:** ask questions until the frame was full then report back
  - More sophisticated: know when to answer questions, when to ask a clarification question, etc. -> Dialog Acts

# Natural Language Understanding

---

- **Classic:**
  - Meaning = Syntactic -> Semantic -> Pragmatic
- **Modern, most used in DS:**
  - Simpler Semantic: **GUS** Slot/Filler approach
  - It assumes a **FRAME** semantics



# Frame-based dialogue agents

---

- Sometimes called "task-based dialogue agents"
- Systems that have the goal of helping a user solve a task like making a travel reservation or buying a product

## Architecture:

- First proposed in the GUS system of 1977. A knowledge structure representing user intentions. One or more frames, each consisting of slots with values
- 3 Step-Algorithm:
  1. Domain detection -> Set of Frames
  2. Intent Detection -> a Frame
  3. Slot Filling -> Filled Frame (entities)

***"It's more than 4 decades years old, but still used in most industrial task-based dialogue agents."***

# The Frame

---

- A set of **slots**, to be filled with information of a given **type**
- Each associated with a **question** to the user
- Sometimes called a **domain ontology**

| Slot     | Type | Question                             |
|----------|------|--------------------------------------|
| ORIGIN   | city | "What city are you leaving from?"    |
| DEST     | city | "Where are you going?"               |
| DEP DATE | date | "What day would you like to leave?"  |
| DEP TIME | time | "What time would you like to leave?" |
| AIRLINE  | line | "What is your preferred airline?"    |

# GUS

GUS: Hello. My name is GUS. I can help you plan a simple trip by air. Where do you want to go ?

Client: I want to go to San Diego on May 28

GUS: What time do you want to leave?

Client: I must be in San Diego before 10 am

GUS: Would you like P.S.A. flight 102 that arrives at 9.15 am ?

Client: What is the next flight ?

GUS: Air California flight 310 that leaves at 8.30am and arrives at 10.00am

Client: I'll take the first one

GUS: In what name should I make a reservation, please?

Client: Dan Bobrow

GUS: I have confirmed the following flight: P.S.A. flight 102 on Wednesday May 28 from San Jose to San Diego that leaves at 7.30 am and arrives at 9.15 am.

GUS: What date do you want to return on ?

Client: On Friday in the evening.

GUS: Would you like the flight that leaves at 7.45 pm ?

Client: That's fine.

GUS: I have confirmed the following flight: P.S.A. flight 307 on Friday May 30 from San Diego to San Jose that leaves at 7.45 pm and arrives at 9.30 pm Thank you for calling. Goodbye

# Control structure for GUS frame architecture

---

- System asks questions of user, filling any slots that user specifies
- User might fill many slots at a time:
  - I want a flight from San Francisco to Denver one way leaving after five p.m. on Tuesday.
- When frame is filled, do database query

# GUS slots have condition-action rules attached

---

Some rules attached to the DESTINATION slot for the plane booking frame:

1. Once the user has specified the destination ->
  - Enter that city as the default *StayLocation* for the hotel booking frame.
2. Once the user has specified DESTINATION DAY for a short trip ->
  - Automatically copy as ARRIVAL DAY.

# GUS systems have multiple frames

---

Frames like:

- Car or hotel reservations
- General route information
  - *Which airlines fly from Boston to San Francisco?,*
- Information about airfare practices
  - *Do I have to stay a specific number of days to get a decent airfare?).*

Frame detection task:

- System must detect which slot of which frame user is filling
- And switch dialogue control to that frame.

# GUS: Natural Language Understanding for filling dialog slots

---

## 1. Domain classification

Asking weather? Booking a flight? Programming alarm clock?

## 2. Intent Determination

Find a Movie, Show Flight, Remove Calendar Appt

## 3. Slot Filling

Extract the actual slots and fillers

# Natural Language Understanding for filling slots

---

Show me morning flights from Boston to SF on Tuesday.

|              |               |
|--------------|---------------|
| DOMAIN:      | AIR-TRAVEL    |
| INTENT:      | SHOW-FLIGHTS  |
| ORIGIN-CITY: | Boston        |
| ORIGIN-DATE: | Tuesday       |
| ORIGIN-TIME: | morning       |
| DEST-CITY:   | San Francisco |



# Natural Language Understanding for filling slots

---

Wake me tomorrow at six.

**DOMAIN: ALARM-CLOCK**

**INTENT: SET-ALARM**

**TIME: 2017-07-01 0600-0800**

# NLU 1. Rule-based Slot-filling

---

Write regular expressions or grammar rules

Wake me (up) | set (the|an) alarm | get me  
up

Note: Really different from Eliza?

# NLU 1. Rule-based Slot-filling

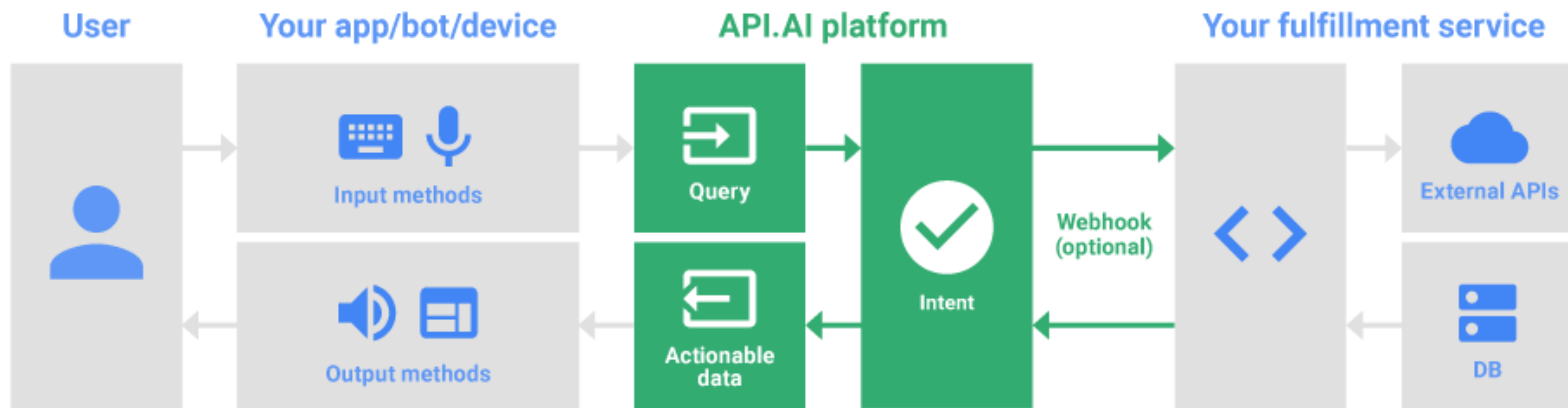
---

- Like many rule-based approaches
- Positives:
  - High precision
  - Can provide coverage if the domain is narrow
- Negatives:
  - Can be expensive and slow to create rules
  - Can suffer from recall problems (**brittle**)

# NLU 2

<https://dialogflow.com/docs>

- Domain
- Intents ~ high-level speech acts
- Entities ~ ontology elements



- Machine Learning -> more is better!
- 1K annotated sentences

# Problems of Frames

---

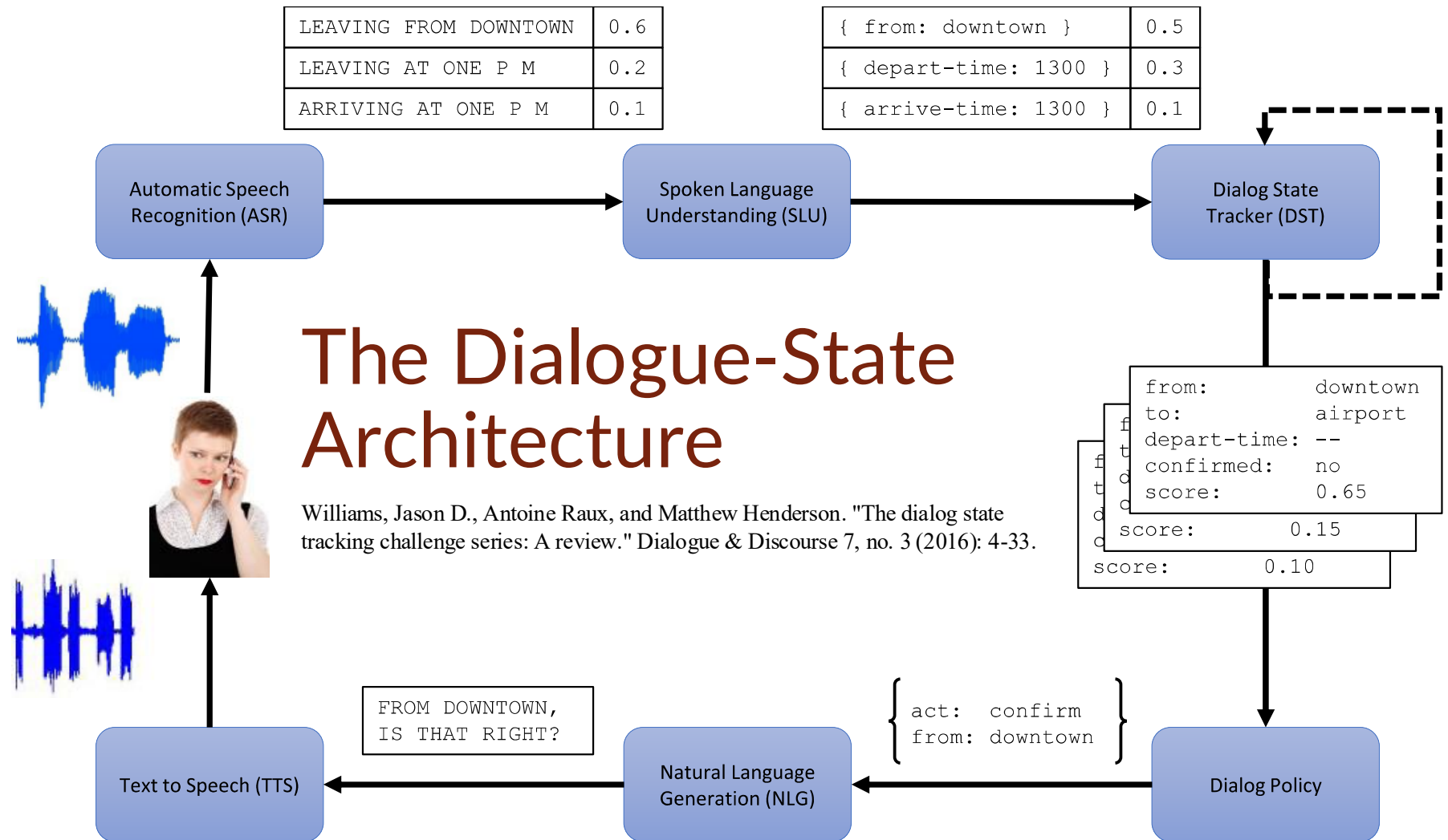
- Not easily applicable to complex task
  - May not be a single frame
  - Dynamic construction of frames

## -> Travel Plans

- Goal: get from Paris to Berlin
- Options: fly, train, drive
- Flight: airline, airport, price, date
- Train: station, class, discount?, reservation?
- Drive: directions, fastest or cheapest?
- Complex Frame
  - Frame hierarchy
  - Agenda

# The Dialogue-State Architecture

Williams, Jason D., Antoine Raux, and Matthew Henderson. "The dialog state tracking challenge series: A review." *Dialogue & Discourse* 7, no. 3 (2016): 4-33.



# Components in a dialogue-state architecture

---

- **NLU:** extracts slot fillers from the user's utterance using machine learning
- **Dialogue state tracker:** maintains the **current state of the dialogue** (user's most recent dialogue act, set of slot-filler constraints from user, ...)
- **Dialogue policy:** decides what the system should do or say next
  - GUS policy: ask questions until the frame was full then report back
  - **More sophisticated:** know when to answer questions, when to ask a clarification question, etc. -> **Dialog Acts**

# Dialogue Acts

- Combine the ideas of **speech acts** and **grounding** into a single representation

Young et al., 2010:

| Tag                              | Sys | User | Description                                                |
|----------------------------------|-----|------|------------------------------------------------------------|
| HELLO( $a = x, b = y, \dots$ )   | ✓   | ✓    | Open a dialogue and give info $a = x, b = y, \dots$        |
| INFORM( $a = x, b = y, \dots$ )  | ✓   | ✓    | Give info $a = x, b = y, \dots$                            |
| REQUEST( $a, b = x, \dots$ )     | ✓   | ✓    | Request value for a given $b = x, \dots$                   |
| REQALTS( $a = x, \dots$ )        | ✗   | ✓    | Request alternative with $a = x, \dots$                    |
| CONFIRM( $a = x, b = y, \dots$ ) | ✓   | ✓    | Explicitly confirm $a = x, b = y, \dots$                   |
| CONFREQ( $a = x, \dots, d$ )     | ✓   | ✗    | Implicitly confirm $a = x, \dots$ and request value of $d$ |
| SELECT( $a = x, a = y$ )         | ✓   | ✗    | Implicitly confirm $a = x, \dots$ and request value of $a$ |
| AFFIRM( $a = x, b = y, \dots$ )  | ✓   | ✓    | Affirm and give further info $a = x, b = y, \dots$         |
| NEGATE( $a = x$ )                | ✗   | ✓    | Negate and give corrected value $a = x$                    |
| DENY( $a = x$ )                  | ✗   | ✓    | Deny that $a = x$                                          |
| BYE()                            | ✓   | ✓    | Close a dialogue                                           |



# Dialogue Acts

Young et al., 2010:

| Utterance                                                           | Dialogue act                                                                         |
|---------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| U: Hi, I am looking for somewhere to eat.                           | <code>hello(task = find, type=restaurant)</code>                                     |
| S: You are looking for a restaurant. What type of food do you like? | <code>confreq(type = restaurant, food)</code>                                        |
| U: I'd like an Italian somewhere near the museum.                   | <code>inform(food = Italian, near=museum)</code>                                     |
| S: Roma is a nice Italian restaurant near the museum.               | <code>inform(name = "Roma", type = restaurant, food = Italian, near = museum)</code> |
| U: Is it reasonably priced?                                         | <code>confirm(pricerange = moderate)</code>                                          |
| S: Yes, Roma is in the moderate price range.                        | <code>affirm(name = "Roma", pricerange = moderate)</code>                            |
| U: What is the phone number?                                        | <code>request(phone)</code>                                                          |
| S: The number of Roma is 385456.                                    | <code>inform(name = "Roma", phone = "385456")</code>                                 |
| U: Ok, thank you goodbye.                                           | <code>bye()</code>                                                                   |

# Dialogue state tracking

User: I'm looking for a cheaper restaurant  
inform(price=cheap)  
System: Sure. What kind - and where?  
User: Thai food, somewhere downtown  
inform(price=cheap, food=Thai, area=centre)  
System: The House serves cheap Thai food  
User: Where is it?  
inform(price=cheap, food=Thai, area=centre); request(address)  
System: The House is at 106 Regent Street

**Figure 15.8** The output of the dialogue state tracker after each turn ([Mrkšić et al., 2017](#)).

# Dialogue state tracker/manager

---

I'd like Cantonese food near the Mission district.



```
inform(food=cantonese, area=mission)
```

1. Dialogue act interpretation algorithm:
  - supervised classification to choose inform based on encodings of current sentence + prior dialogue acts
2. Simple dialogue state tracker:
  - Run a slot-filler after each sentence

# Outline

---

- Introduzione allo studio computazionale del dialogo
- Architetture per ChatBot
- Architetture per Dialog Systems
  - Dialog State and Dialogue Manager
- Evaluation

# Trindi Ticklist

---

**Q1:** *Is utterance interpretation sensitive to context?*

**Q2:** *Can the system deal with answers to questions that give more information than was requested?*

**Q3:** *Can the system deal with answers to questions that give different information than was actually requested ?*

**Q4:** *Can the system deal with answers to questions that give less information than was actually requested?*

**Q5:** *Can the system deal with ambiguous designators?*

**Q6:** *Can the system deal with negatively specified information?*

**Q7:** *Can the system deal with no answer to a question at all?*

...

<https://gup.ub.gu.se/file/207630>

# Trindi Ticklist

---

...

**Q8:** *Can the system deal with noisy input?*

**Q9:** *Can the system deal with 'help' sub-dialogues initiated by the user?*

**Q10:** *Can the system deal with 'non-help' sub-dialogues initiated by the user?*

**Q11:** *Does the system only ask appropriate follow-up questions?*

**Q12:** *Can the system deal with inconsistent information?*

**NEW->**

**Q13:** *Can the system deal with belief revision?*

**Q14:** *Can the system deal with barge-in input?*

**Q15:** *Is it possible to get a system tutorial concerning the kind of information the system can provide and what the constraints on input are?*

**Q16:** *Does the system check its understanding of the user's utterances?*

# A recent ticklist application

LARSSON, S.: User-initiated Sub-dialogues in State-of-the-art Dialogue Systems. Proceedings of the 18th Annual SIGdial Meeting on Discourse and Dialogue, 15-17 August 2017, Saarbrücken, Germany, 2017.

|                                                 | Siri | API.AI | Houndify | Cortana | Alexa |
|-------------------------------------------------|------|--------|----------|---------|-------|
| F1. User anytime jump to task within app        | +    | +      | ?        | -       | +     |
| F2. System resume after within-app sub-dialogue | -    | -      | ?        | -       | -     |
| F3. Signal task resumption                      | N/A  | N/A    | ?        | -       | N/A   |
| F4. User anytime jump across apps               | +    | -      | -        | -       | (+)   |
| F5. System resume after other-app subdialog     | -    | N/A    | N/A      | N/A     | -     |
| F6. Signal app (and task) resumption            | N/A  | N/A    | N/A      | N/A     | N/A   |

Table 1: Dialogue behaviours related to user-initiated sub-dialogues in five dialogue systems

# Evaluating chatbots and task-based dialogue

---

- Chatbots: mainly by human evaluation
  - <https://www.amazon.science/alexaprize>
- Task-based dialogue: mainly by measuring task performance



# Chatbots are evaluated by humans

---

- **Participant evaluation:** The human who talked to the chatbot assigns a score
- **Observer evaluation:** third party who reads a transcript of a human/chatbot conversation assigns a score.

# Observer evaluation: acute-eval

---

Annotators look at two conversations (A + B) and decide which is better:

- **Engagingness:** Who would you prefer to talk to for a long conversation?
- **Interestingness:** If you had to say one of these speakers is interesting and one is boring, who would you say is more interesting?
- **Humanness:** Which speaker sounds more human?
- **Knowledgeable:** If you had to say that one speaker is more knowledgeable and one is more ignorant, who is more knowledgeable?

# The ACUTE-EVAL method, Li et al., 2019

|                                                                                             |                                                                                    |
|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Hi how are you today                                                                        | Hello there, how are you?                                                          |
| Hi, I'm pretty good! Just listening to some aerosmith, they're my fave :) whatre you up to? | I am doing great. How are you?                                                     |
| I am listening to some italian music                                                        | I am great, I did something crazy for me and colored my hair blue!                 |
| Italian music, nice! What do you do for work?                                               | I have a daughter and a son who also love blue colored balls. You should meet them |
| I'm a professional wrestler                                                                 | Well that neat, I got a new car my mother gave so maybe I could see them!          |
| Oh wow! I'm a construction worker, I've been doing it since I was 16.                       | It is a beautiful city. And, I try to be... Just cannot afford a bigger house atm. |
| What do you do for a living                                                                 | I am sorry to hear that, I feel bad going out of town for spring break now.        |
| I'm a construction worker haha, just told you! Whats your favorite type of food?            | Ok. I going to school in the spring for casino manager                             |
| My favorite is garlic bread                                                                 | Well I turn 29 next week, I wonder if that is a good age to apply as one.          |
| Yum! I love mexican food, so good :) do you have any hobbies?                               | My grandmother just died from lung cancer, sucks                                   |
| I like to workout and go to the gym                                                         |                                                                                    |
| We're a bit different- I love watching nascar and ufc. They're so fun!                      |                                                                                    |

**Who would you prefer to talk to for a long conversation?**

☐ I would prefer to talk to **Speaker 1**    ☐ I would prefer to talk to **Speaker 2**

**Please provide a brief justification for your choice (a few words or a sentence)**

Please enter here...

Figure from Li, M., Weston, J., and Roller, S. (2019). Acute-eval: Improved dialogue evaluation with optimized questions and multi-turn comparisons. *NeurIPS19 Workshop on Conversational AI*.

# Task-based systems are evaluated by task success!

---

1. End-to-end evaluation (Task Success)

2. Slot Error Rate for a Sentence:

# of inserted/deleted/substituted slots

---

# of total reference slots for sentence

# Evaluation Metrics: Slot error rate

- *Make an appointment with Chris at 10:30 in Gates 104*

| Slot   | Filler     |
|--------|------------|
| PERSON | Chris      |
| TIME   | 11:30 a.m. |
| ROOM   | Gates 104  |

**Slot error rate: 1/3**

**Task success:** At end, was the correct meeting added to the calendar?

# More fine-grained metrics: User Satisfaction Survey

Walker, Marilyn, Candace Kamm, and Diane Litman. "Towards developing general models of usability with PARADISE." *Natural Language Engineering* 6, no. 3 & 4 (2000): 363-377.

|                          |                                                             |
|--------------------------|-------------------------------------------------------------|
| <b>TTS Performance</b>   | Was the system easy to understand ?                         |
| <b>ASR Performance</b>   | Did the system understand what you said?                    |
| <b>Task Ease</b>         | Was it easy to find the message/flight/train you wanted?    |
| <b>Interaction Pace</b>  | Was the pace of interaction with the system appropriate?    |
| <b>User Expertise</b>    | Did you know what you could say at each point?              |
| <b>System Response</b>   | How often was the system sluggish and slow to reply to you? |
| <b>Expected Behavior</b> | Did the system work the way you expected it to?             |
| <b>Future Use</b>        | Do you think you'd use the system in the future?            |

# Other Heuristics

---

## Efficiency cost:

- total elapsed time for the dialogue in seconds,
- the number of total turns or of system turns
- total number of queries
- “turn correction ratio”: % of turns that were used to correct errors

## Quality cost:

- number of ASR rejection prompts.
- number of times the user had to barge in